① $(\Omega, \mathcal{F}, P)$ $\Omega = \{(\omega_0, \omega_1),\ \omega_i \in \{-1, 0, 1\}\}$

$\mathcal{F} = 2^{\Omega}$

$P(\omega_i = 0) = \frac{1}{2}$ $P(\omega_i = \pm 1) = \frac{1}{4}$

$X_0 = \omega_0$ $X_1 = \omega_0 \omega_1$ $P(X_1)$, $E(X_0)$, $\sigma(X_0)$, $\sigma(X_1)$

$E[X_1 | X_0]$

$P(X_1 = 0) = P(\omega_1 = 0$ или $\omega_0 = 0) = 1 - P(\omega_1 \neq 0, \omega_0 \neq 0) = 1 - \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4}$

$P(X_1 = 1) = P(\omega_0 = \omega_1 = \pm 1) = 2 \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{8}$

$P(X_1 = -1) = P(\omega_0 = -\omega_1 = \pm 1) = 2 \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{8}$

$E[X_0] = (-1) \cdot \frac{1}{4} + 0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{4} = 0$

$\sigma(X_0)$: $A_0 = \{\omega_0 = 0\}$ → $\{(0,1)(0,0)(0,-1)\}$ $A_{-1} = \{\omega_0 = -1\}$

$A_1 = \{\omega_0 = 1\}$ → $\{(1,1)(1,0)(1,-1)\}$

$\sigma(X_1) = \sigma\{\{(\omega_0 \omega_1)$: $\omega_0 = 0$ или $\omega_1 = 0\} =$

$= \{(0,1), (0,-1), (1,0), (-1,0), (0,0)\}$

$\{(-1,1), (1,-1)\}$

$\{(1,1), (-1,-1)\}\}$

$E[X_1 | X_0] = E[\omega_0 \omega_1 | \omega_0] =$

$$E[X_1 | \overset{\sigma(X_0)}{\mathcal{F}}] = \sum E[X | A_i] I_{A_i} = \cancel{E[X_1 | \{(0,\cdot)\}]} \overset{0}{I}_{\{(0,\cdot)\}} + \cancel{E[X_1 | \{(1,\cdot)\}]} I_{\{(1,\cdot)\}} + \cancel{E[X_1 | \{(-1,\cdot)\}]} I_{\{(-1,\cdot)\}} = 0$$

0 $X_1((1,0)) \cdot P(\omega_1 = 0) + X_1((1,1)) P(\omega_1 = 1) + X_1(1,-1) \cdot P(\omega_1 = -1)$ → 0 → 0

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② $X_1 \dots X_n \dots$ – iid $P(X_i = 1) = p$

$P(X_i = -1) = q = 1 - p$

$\mathcal{F}_n = \sigma\{X_i,\ i = \overline{1, n}\}$

$S_n = \sum_{i=1} X_i$

① Док-ть $M_n = \left(\frac{q}{p}\right)^{S_n}$ – мартингал

② Если $p = q = \frac{1}{2}$ доказ. S_n – мартингал, $S_n^2 - n$ – мартингал

M_n – мар. если $\boxed{E(M_n | \mathcal{F}_m) = M_m \quad m < n}$

$\boxed{E(M_n | \mathcal{F}_{n-1}) = M_{n-1}}$

$\mathcal{F}_{n-2} \subset \mathcal{F}_{n-1}$

$E(M_n / \mathcal{F}_{n-2}) = E(E(M_n / \mathcal{F}_{n-1}) / \mathcal{F}_{n-2}) =$

$= E(M_{n-1} / \mathcal{F}_{n-2}) = M_{n-2}$

① $E(M_{n+1} | \mathcal{F}_n) = E\left(\left(\frac{q}{p}\right)^{S_{n+1}} | \mathcal{F}_n\right) = E\left(\left(\frac{q}{p}\right)^{S_n + X_{n+1}} | \mathcal{F}_n\right) =$

$= E\left(M_n \cdot \left(\frac{q}{p}\right)^{X_{n+1}} / \mathcal{F}_n\right) = M_n E\left[\left(\frac{q}{p}\right)^{X_{n+1}} | \mathcal{F}_n\right] =$ → $X_1 \dots X_n$

$= M_n \cdot E\left[\left(\frac{q}{p}\right)^{X_{n+1}}\right] = M_n$ $\Rightarrow M_n$ – мартингал

$\left(\frac{q}{\cancel{p}}\right)^1 \cdot \cancel{p} + \left(\frac{q}{p}\right)^{-1} \cdot q = q + p = 1$

② Д-ть S_n мартин. $p = q = \frac{1}{2}$

$S_n^2 - n$ – мартин.

$$E(S_{n+1} | \mathcal{F}_n) = E(S_n + X_{n+1} | \mathcal{F}_n) = \underset{S_n}{E(S_n | \mathcal{F}_n)} + \underset{E(X_{n+1}) = 0}{E(X_{n+1} | \mathcal{F}_n)} = S_n$$

$1 \cdot p + (-1) \cdot q = q - p$

$$S_n^2 - n \overset{?}{=} E(S_{n+1}^2 - (n+1) | \mathcal{F}_n) = E(S_{n+1}^2 | \mathcal{F}_n) - E(n+1 | \mathcal{F}_n) = E(S_{n+1}^2 | \mathcal{F}_n) - (n+1) =$$

$$= E((S_n + X_{n+1})^2 | \mathcal{F}_n) - (n+1) = \underset{S_n^2}{E(S_n^2 | \mathcal{F}_n)} + 2\underset{2S_n \cdot E(X_{n+1} | \mathcal{F}_n) = 0}{E(S_n \cdot X_{n+1} | \mathcal{F}_n)} + \underset{1}{E(X_{n+1}^2 | \mathcal{F}_n)} - (n+1) = S_n^2 - n$$

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③ $X_i = +1$ $p = \frac{1}{2}$ $X_i = -1$ $p = \frac{1}{2}$ $S_n = \sum X_i$

Останав. если $S_n = a$ либо $S_n = -b$ → случ. блуждания

① $P(S_n = a)$ когда ост.

② $E(T)$ – кол. раунд. до ост.

$T_+ = \inf\{n > 0,\ S_n = a\}$ $T = \min\{T_+, T_-\}$

$T_- = \inf\{n > 0,\ S_n = -b\}$

$T = T_+$ → выигр

$T = T_-$ → проигр

$P(T = T_+) = ?$

Optional Stopping Theorem

M_i – мартингал

$E[M_\tau] = ~~E~~[M_0]$

τ – время останов.

$E[M_{n}] = E[M_0]$

$E[M_n | \Omega]$

1) Если $\tau < K < \infty$

2) Если $|M_n| < K$

3) Если $|M_n - M_{n-1}| < K$

$|S_{n+1} - S_n| = 1$ $\Rightarrow$ $E[S_T] = E[S_0] = 0$

①
$$\begin{cases} a \cdot P(T = T_+) - b\,P(T = T_-) = 0 \\ P(T = T_+) + P(T = T_-) = 1 \end{cases}$$

$P(T = T_+) = \frac{b}{a} P(T = T_-)$

$\left(1 + \frac{b}{a}\right) P(T = T_-) = 1$

$\frac{a+b}{a}$

$P(T = T_+) = \frac{b}{a+b}$

$P(T = T_-) = \frac{a}{a+b}$

② Матожид. $E[T]$ $E[S_T] = E[S_0]$?

$M_n = S_n^2 - n$ – мартингал

$0 = E[M_0] = E[M_T] = E[S_T^2 - T] = E[S_T^2] - E[T]$

$$E[T] = E[S_T^2] = b^2 P(T = T_-) + a^2 P(T = T_+) =$$

$$= \frac{b^2 \cdot a}{a+b} + \frac{a^2 \cdot b}{a+b} = \frac{ab(a+b)}{a+b} = ab$$

$E[T] = ab$

④ p, q $p \neq q$ S_n – не марг.

$M_n = \left(\frac{q}{p}\right)^{S_n}$ → мартингал